COCCON_NEW: Proactive Emotional Support Across Sessions

Grounded reasoning, safety checks, and need-state memory

Arpit Bhomia (12340340) · Ashish Ranjan (12340370)

Extends COCOON: Fu et al., EMNLP 2025





Stock photo, illustrative

What is the real problem?

People show a feeling, not the need behind it. Pushing too hard can cause resistance.

WHAT WAS SAID

“My boss gave me more feedback on my project. Feels like they don't see all the hard work I put in.”

the need is never stated

THE NEED UNDERNEATH (hidden from the supporter)

“I need to feel valued and recognized for my contributions.”

RISK OF PUSHING TOO FAR

“Can you tell me more about the specifics of the campaign and what you were responsible for?”

Every supporter reply in this saved session asked for more detail.

Excerpt: saved test conversation p000852, session 1 (details in notes)

Earlier work and the original COCOON method

Reactive support

ESConv · ExTES

Person asks for help → **States the problem** → **Supporter picks a strategy** from a fixed list → **Reply**

ESConv: English, 1.3k dialogues, 8 strategies · ExTES: English, 11.2k dialogues, 16 strategies (LLM-generated)

COCOON

Fu et al., EMNLP 2025

Active listening: infer the need behind the feeling; reflect rather than question

User profile emotion · feeling · need · memory → **Two-stage dialogues** listen, then suggest → **AELS filter** drop low-quality → **Annotate** Analysis + Strategy per turn → **Fine-tune** LLaMA-3-8B-Instruct → **Evaluate** simulated users, scales

3.5k dialogues · Chinese · 18.82 turns on average · generated and annotated with GPT-4o

Our setting: COCOON's corpus and pipeline are unreleased, so we rebuilt the method in English with open-weight models.

Sources: Fu et al. (EMNLP 2025) Table 1, Sec. 3, Fig. 2, App. F; SOP Sec. I and IV-A

Gaps we targeted and our four additions

GAPS		OUR ADDITIONS (SOP ENHANCEMENTS)
No memory across sessions Each dialogue stands alone *partly noted by the COCOON authors*	*continuity across sessions* → E1	**Linked English sessions** 2–4 sessions per person, gaps of 1–56 days
	continuity across sessions → E4	**Persistent need-state memory** Needs kept as beliefs that can change
No check before a reply is sent Analysis, strategy and reply in one pass *identified in our SOP*	*inspection before release* → E2	**Four grounded agents + calibrated gate** Analyzer · Strategist · Generator · Critic
Resistance measured only in total No per-turn intrusiveness or resistance score *identified in our SOP*	*measurement of resistance* → E3	**Turn-level IP and PRI** Intrusiveness of each reply; resistance in the next

Supporting work (not extra enhancements): QLoRA fine-tuning at 0.5B / 3B / 7B · steering and DPO comparisons · reliability checks

The corpus we built

1,000
seeds
EmpatheticDialogues situations

→

1,000
profiles
hidden need chain, persona, resistance level

→

2,493
sessions generated
for 820 train/val profiles; Qwen2.5-7B plays both sides

→

2,369
sessions kept
from 801 profiles; 66 rejected, 58 later sessions dropped

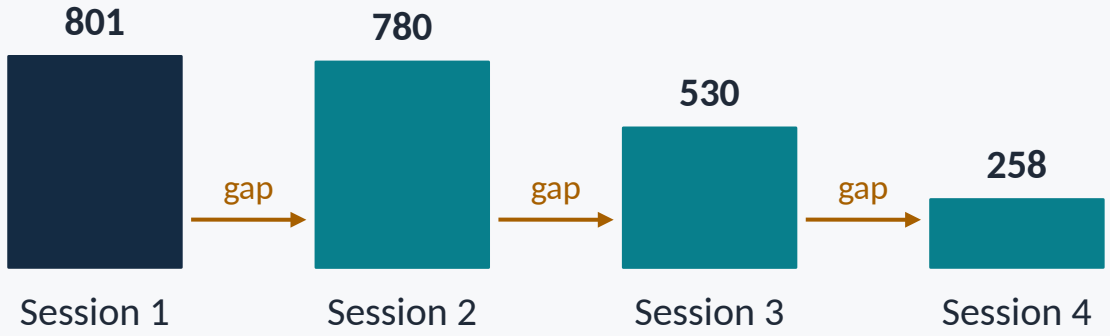
→

21,429
annotated turns
Analysis + Strategy for every supporter turn

Filter: broken structure, leaked need, non-English turns, AELS listening score below 5 (Mistral-Nemo judge); crisis terms go to quarantine

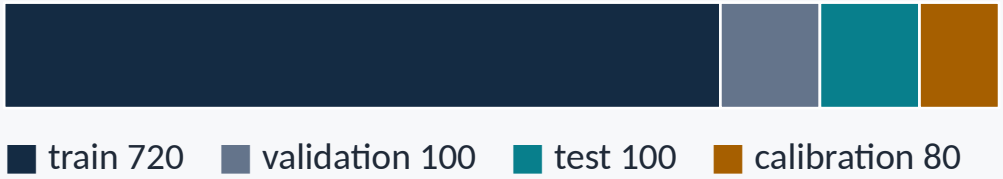
Linked sessions per person

Between sessions: 1–56 days (median 7.6); the need resolves, intensifies or shifts



sessions kept at each position

How the 1,000 profiles are used

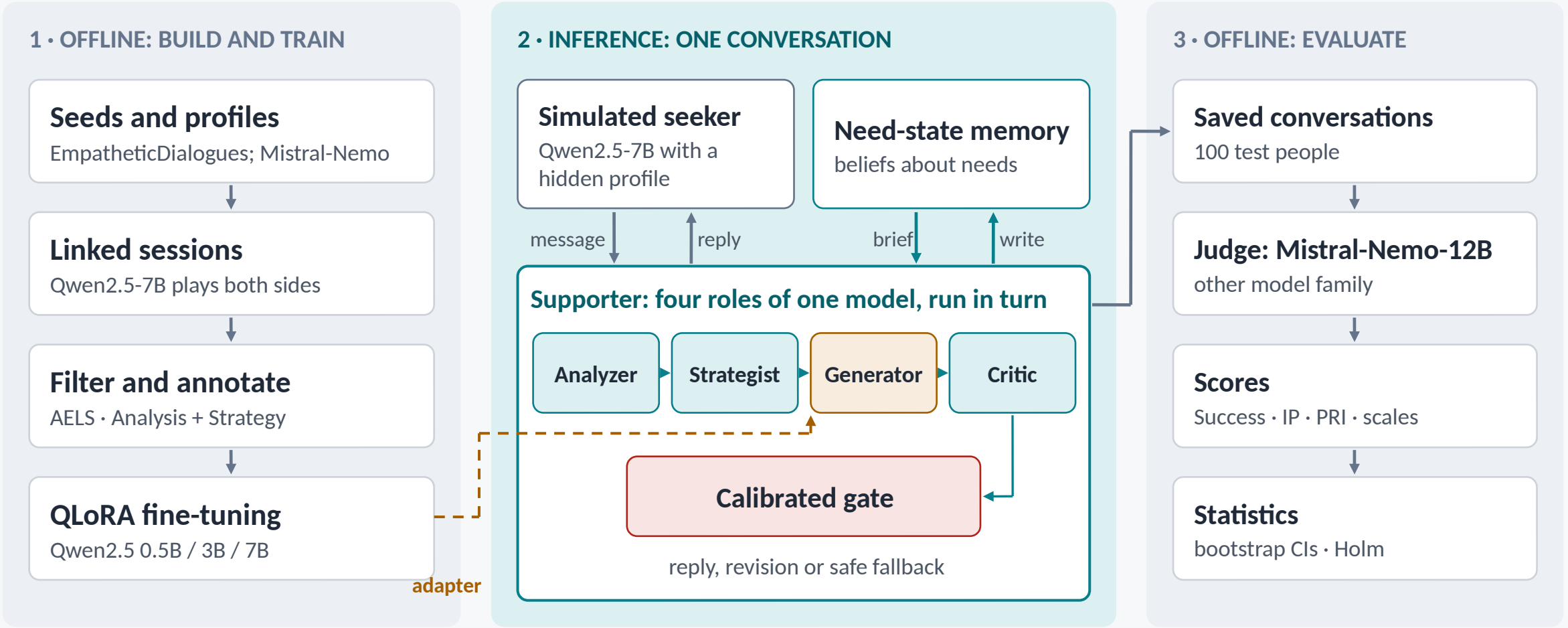


Train + val (820): corpus dialogues; 801 kept at least one session

Test (100) and calibration (80): no corpus dialogues; held out for evaluation and for calibrating the gate

Sources: results/v3_full1000/dataset/dataset_stats.json; data/corpus/*.jsonl; src/filter.py; configs/default.yaml

Architecture of our solution



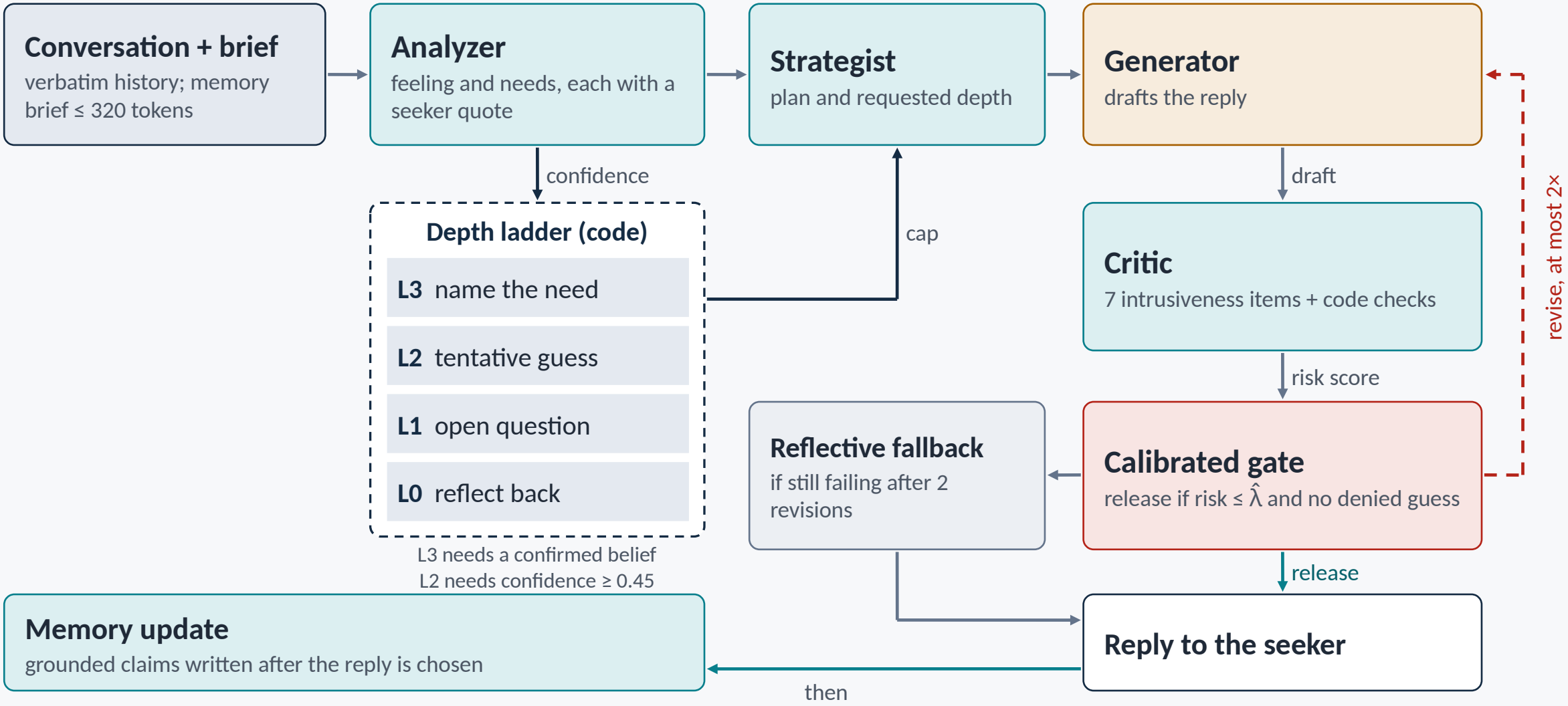
Qwen2.5-7B-Instruct dialogues · 4 agent roles · seeker

Fine-tuned 7B adapter Generator, full system

Mistral-Nemo-12B profiles · filter · annotation · judge

One model on the GPU at a time; the judge loads only after generation ends. Sources: docs/architecture/program-light.svg; src/; configs/models.yaml

How one supporter reply is produced



Sources: src/agents.py (AgentPipeline.step, Critic); src/pacing.py; src/conformal.py; configs/default.yaml; docs/architecture/turn-light.svg

Architecture of need-state memory

1 · EVIDENCE ENTERS

Analyzer claim
need · confidence · seeker quote

↓

Quote check (code)
no word-for-word quote: refused

↓

Route the claim
propose · support · confirm · deny · resolve

written after each reply

2 · BELIEFS ARE STORED

One belief (node)

need	the need, in words
depth · parent	0 (feeling) to 3 · parent need
status	hypothesis · confirmed · denied · resolved
confidence	0 to 1
quotes	seeker's words for and against
time	first / last seen · sessions seen
flags	dormant · re-proposal blocked

Append-only log: every change is a row

3 · A BRIEF IS READ

Brief (≤ 320 tokens)

- best guess and its chain
- up to two rival needs
- links not yet tested
- denied guesses, in the seeker's words
- resolved and faded needs

read before each turn by

Analyzer | **Strategist**

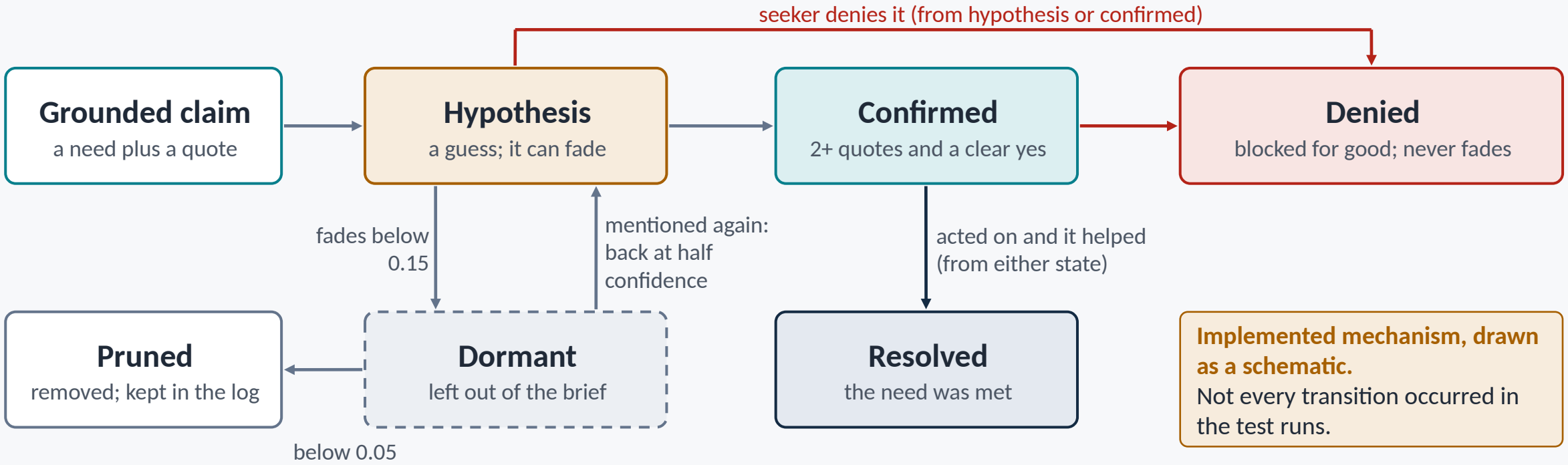
Generator | **Critic**

A belief that can change: a guess about a need that gains support, is confirmed, is withdrawn when denied, or fades with time.

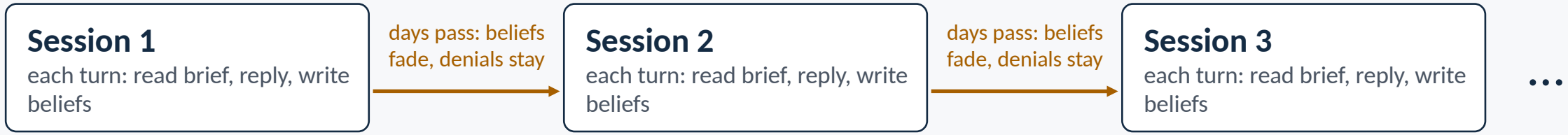
Not connected: the seeker's hidden profile. Only the simulator and the Success judge read it.

Sources: src/memory.py (MemoryNode, NeedStateMemory); src/agents.py (update_memory); src/grounding.py; docs/architecture/memory-light.svg

How a need belief changes across sessions



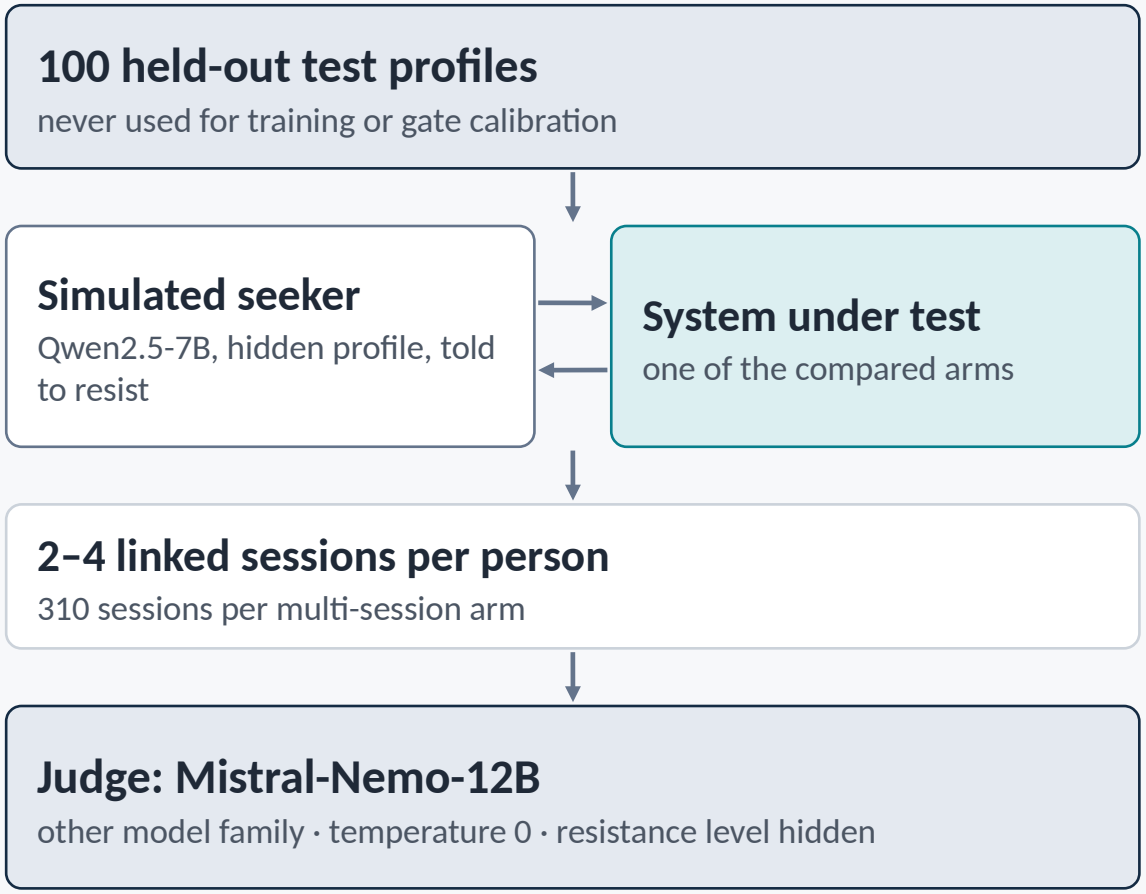
ACROSS SESSIONS (SCHEMATIC)



Confirmed needs fade more slowly, and each later session that brings a need back slows its fading further.

Sources: src/memory.py (confirm, disconfirm, resolve, decay, prune, reinstate, reopen); src/agents.py; src/evaluate.py; configs/default.yaml

How we evaluated the system



Success higher is better
Did the supporter name the hidden need? One 1–7 judge rating per session, rescaled to 0–1.

IP · intrusiveness lower is better
How far a reply oversteps what the seeker was ready for. 7 items; 4 sampled replies per session.

PRI · resistance lower is better
Resistance in the seeker's next reply (observed). Counterfactual PRI: real turn vs neutral reflection, 5 replays each.

95% CI: bootstrap over people · Holm correction over 42 headline contrasts

Seekers are simulated and every score comes from an LLM judge; there are no human ratings.

Sources: prompts/scale_success.md, metric_ip.md, metric_pri.md; src/counterfactual.py; src/judge.py; configs/models.yaml; temp_results/23_multiple_comparisons

Ablation study: what does each component add?

Ablation: switch one part on or off, keep everything else fixed, and compare.

SHARED BY A–D

- **Untuned Qwen2.5-7B**
- **100 test people**
- **2–4 sessions each**
- **Same simulated seeker**

	One call	Four agents
Gate off	**A** one call, no gate	**C** four agents, no gate
Gate on	**B** one call + gate	**D** four agents + gate

red outline = gate on

OTHER CONTROLLED COMPARISONS

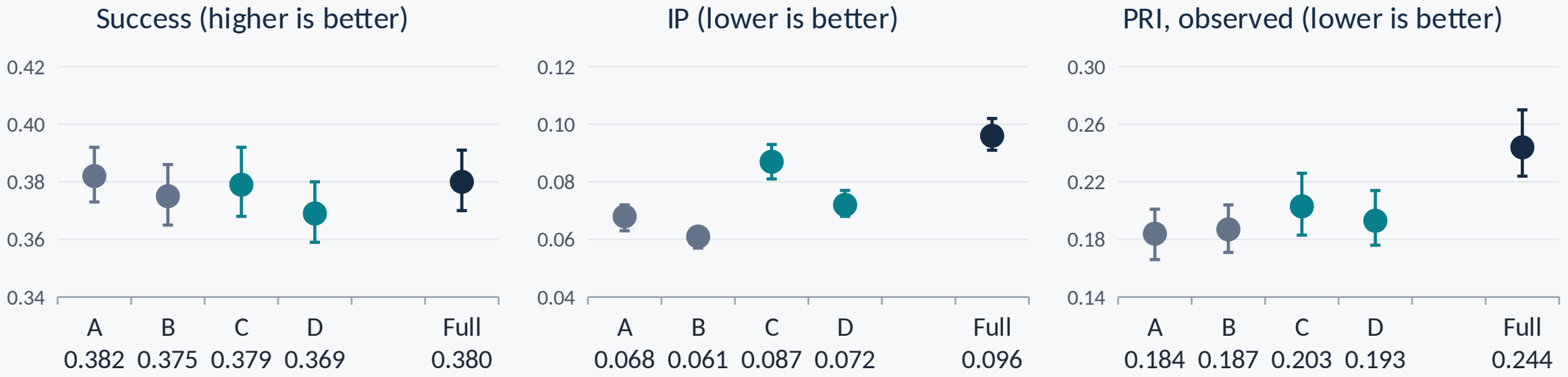
Annotations: with vs without thoughts
fine-tuned 0.5B and 7B, Analysis + Strategy in the target or not

Memory: on vs off, and memory type
untuned: 4 memory types · fine-tuned 7B: with vs without

Sources: configs/arms.yaml (cellA–cellD, sft_*, mem_*, sft7b_multi_*); SOP Sec. III-B; temp_results/06, 05, 13, 07

Results: architecture and full system

A–D: untuned 7B, Ollama · Full system: D + fine-tuned Generator + need-state memory, transformers (backends differ)





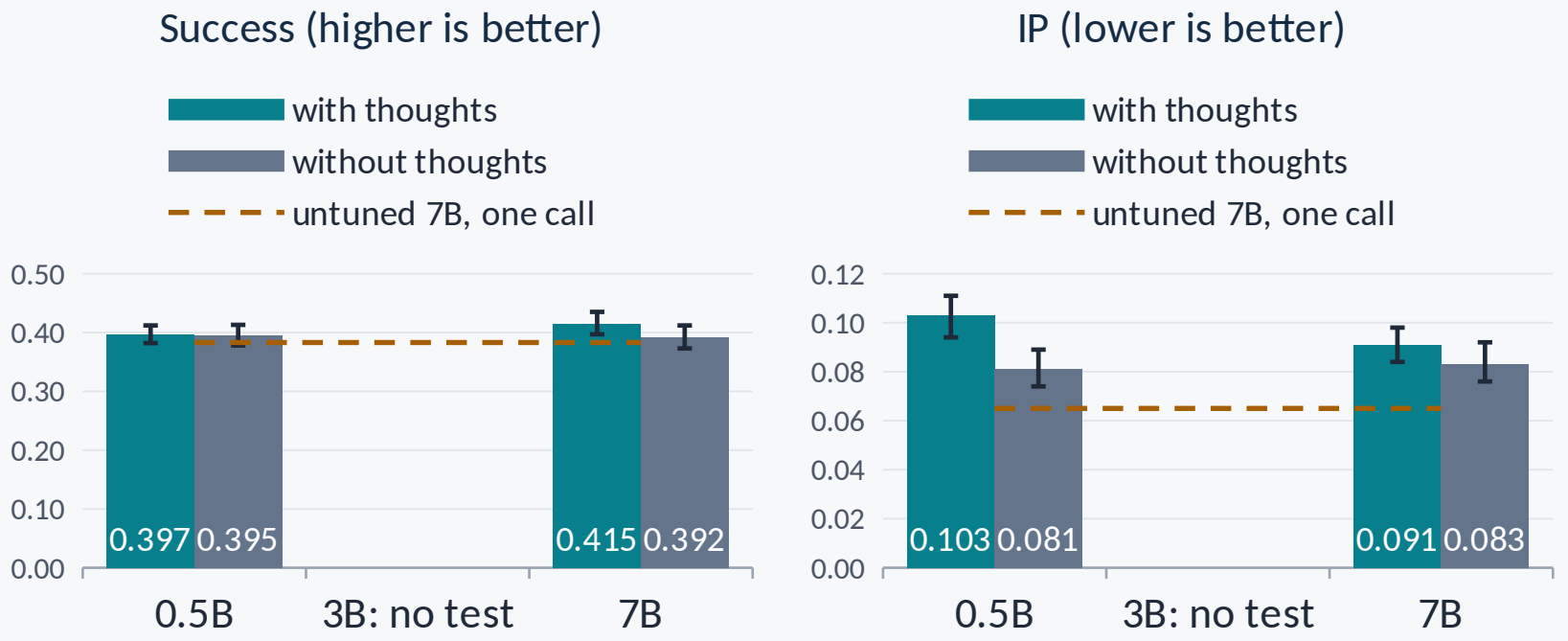
Gate cell	turns	released	fallback
D, our people	1,240	84%	16%
Full, our people	1,240	98%	2%
D, ExTES people (out of distribution)	400	35%	65%

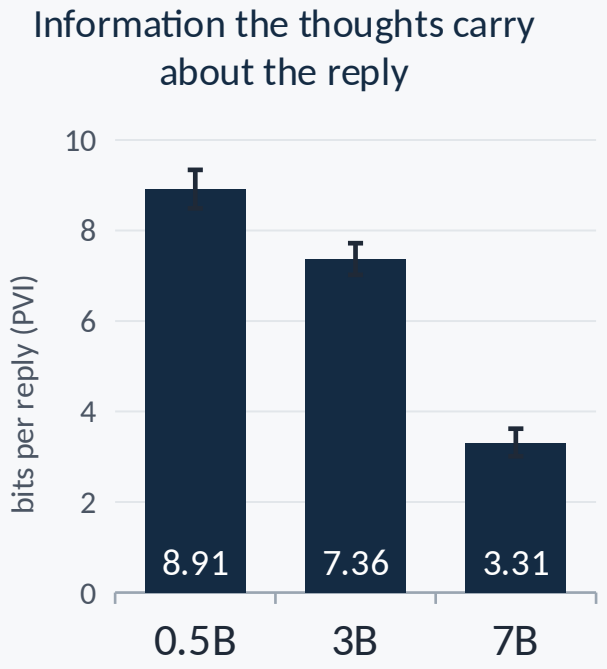
Whiskers: 95% bootstrap CI over people · n = 100 people, 310 sessions per arm · † CI derived (notes)

Sources: temp_results/06 (cells.txt, cellD_scores_complete.txt), 20_step12, 17_gate_violation_rate; results/v3_full1000/baselines_1000, sft_agents

Results: fine-tuning and annotations

Single session · 100 test people · one-call supporter, no gate, no memory · Qwen2.5 + QLoRA





validation turns, n = 2,608 per size

With − without thoughts (95% CI)	Success	IP
0.5B	+0.002 [−0.018, +0.023]	+0.021 [+0.011, +0.032]
7B	+0.023 [0.000, +0.047]	+0.007 [−0.002, +0.017]

PVI: Qwen2.5 log-prob of the gold reply with vs without the gold thoughts

Bars: mean with 95% bootstrap CI over people. Dashed line: untuned 7B. Differences: unadjusted CIs. 3B was trained but not dialogue-tested.

Sources: temp_results/05_single_session_test__finetuned_vs_untuned/scores.txt; 02_finetuning/report.txt; 03_information_in_thoughts__PVI

Results: need-state memory

A · Memory baselines: untuned 7B, four agents, gate off

people change between sessions · 100 people, 310 sessions per arm · unpaired



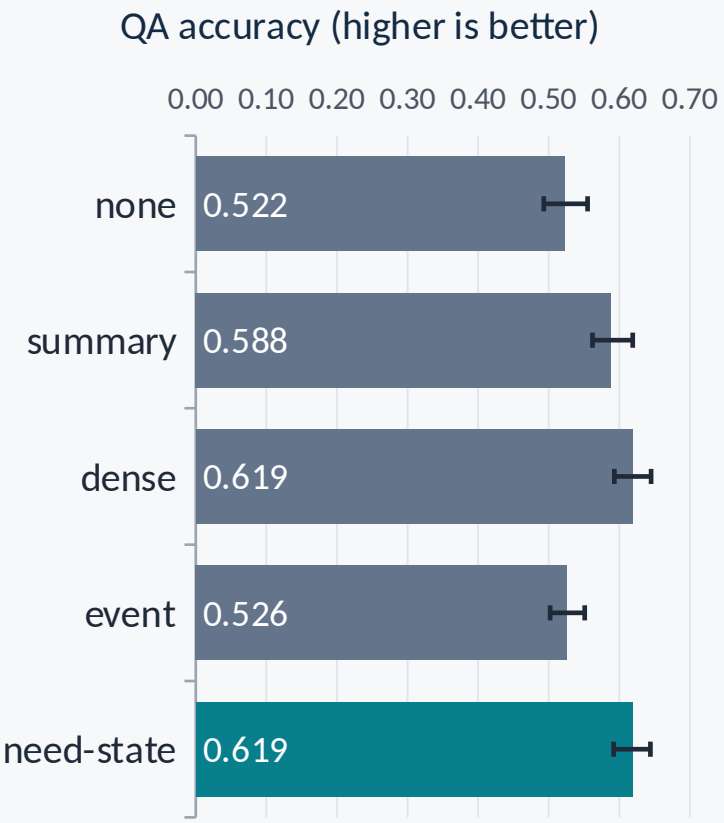
B · Fine-tuned 7B, one call: with vs without need-state memory

people change between sessions · 100 people, 310 sessions per arm · S = session



C · ES-MemEval question answering

external benchmark · 18 users · 1,427 questions each



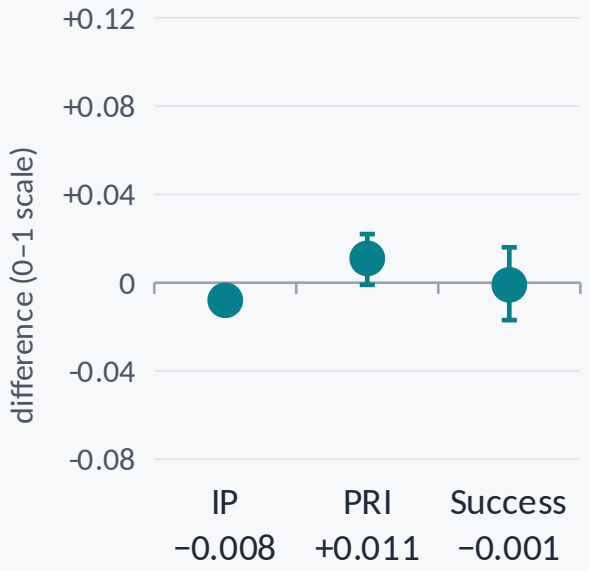
Correct = judge ≥ 5/7 against the gold answer; 95% CI bootstrap over users

Sources: temp_results/13_memory_comparison__untuned_4_arms (memory_arms.txt, es_memeval_report.md); 07_multi_session__finetuned_7B_with_and_without_memory

Results: comparison methods

Activation steering – control

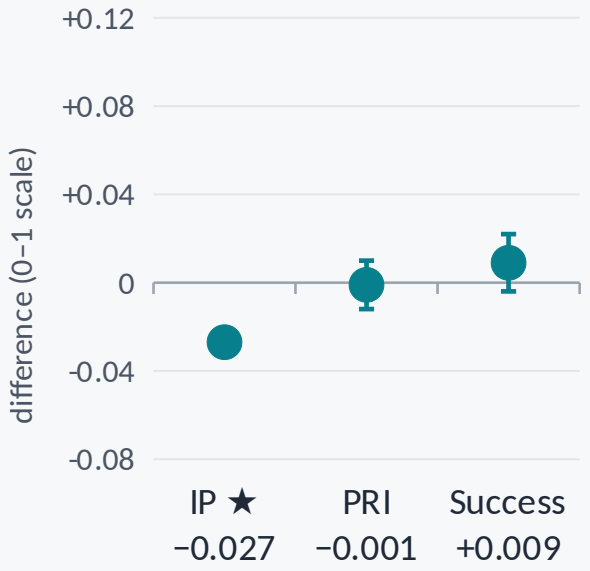
untuned 7B one call · layer 12 × 0.2 vs unsteered · transformers · paired



arm	IP	PRI	Success
control	0.085	0.203	0.405
steered	0.076	0.213	0.404

DPO – its SFT control

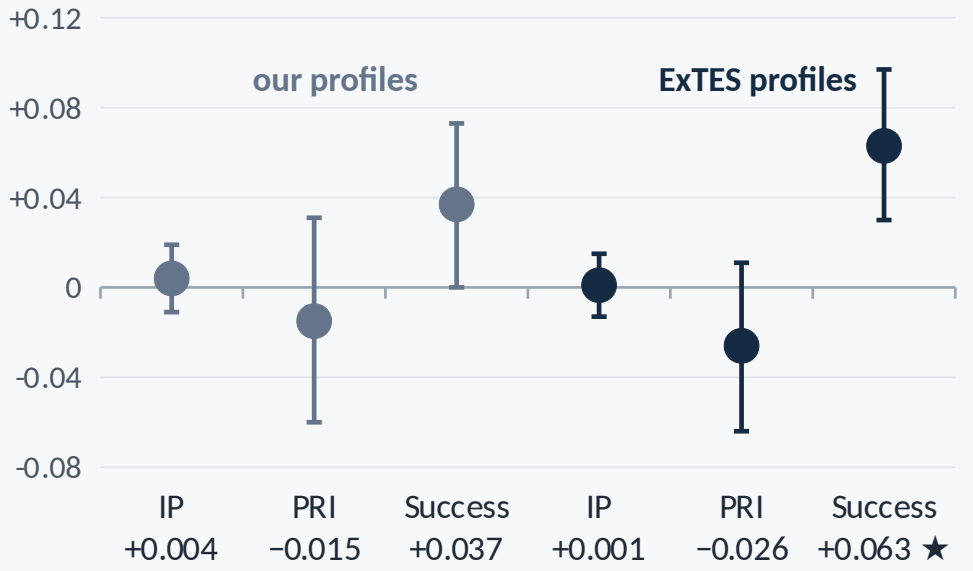
fine-tuned 7B one call · 460 preference pairs · transformers · paired



arm	IP	PRI	Success
SFT control	0.088	0.212	0.377
DPO	0.061	0.211	0.387

ExTES-trained – ours-trained 7B

one session, one call, no gate · 100 people per cell · unpaired · judged on 10.10.10.104



trained on → people	IP	PRI	Success
ours → ours	0.105	0.276	0.390
ExTES → ours	0.109	0.262	0.427
ours → ExTES	0.108	0.320	0.453
ExTES → ExTES	0.109	0.294	0.517

★ survives Holm correction (42 contrasts, α = 0.05) · whiskers: 95% bootstrap CI over people · difference = first arm − second arm; IP and PRI lower is better, Success higher is better

Sources: temp_results/23_multiple_comparisons; 15_activation_steering; 18_preference_alignment_dpo; 16_corpus_comparison__ours_vs_ExTES